

Ebrahim Tahmasebi Boldaji

Naval Architect | CFD Solver Development & Numerical Methods

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Education

Erasmus Mundus Joint Master Degree – Sustainable Ship and Shipping 4.0 <i>Thesis: Machine-Learning Assisted Field Surrogate Models of Complex Ship Propulsion Systems in Hydrodynamic Numerical Simulations</i> Technical University of Hamburg, Germany University of A Coruña, Spain University of Naples Federico II, Italy	Sep. 2024 – Mar. 2026 Grade: 1.7/5 (1:Best, 5:Fail) Grade: 9/10 Grade: 109/110
Master of Science – Naval Architecture Sharif University of Technology, Tehran <i>Thesis: Structural Health Monitoring of Marine Structures with Inverse Finite Element Method</i>	Sep. 2018 – Mar. 2021 GPA: 3.87/4.0
Bachelor of Science – Naval Architecture and Marine Engineering Persian Gulf University, Bushehr <i>Thesis: Investigation of Hydrodynamic Behavior of Planing Craft using Star-CCM+</i>	Sep. 2014 – Jul. 2018 GPA: 3.70/4.0

Technical Skills

Programming	MATLAB, Python, C++
CFD	Star-CCM+, OpenFOAM, FreSCo+, Solution Forcing, Optimization
Solver Development	Custom FVM, FDM, SPH and LBM solver (using Python), fluid-structure interaction via Solid4foam
Structural Analysis	ABAQUS, finite element programming, inverse FEM for structural health monitoring
CAD & Design	SolidWorks, Rhino, AutoCAD, Onshape, Maxsurf
Machine Learning	Surrogate Models (RNN, CNN, CVAE), Reduced Order Models (ROM)
Mathematics	Potential Flow Theory, Linear Stability Analysis, PDEs, Spectral Analysis, Numerical Linear Algebra
Tools	Linux, LaTeX, VS Code, Git/GitHub, Google Colab, Ipe
Domain Knowledge	Ship design (GA, hydrostatics, intact stability, propulsion selection); class regulatory frameworks (SOLAS, MARPOL, IMO, FSICR ice class, CRS scantling); alternative fuels under CII/EEXI/FuelEU

Research Experience

M.Sc. Researcher — ML-Assisted CFD	Sep. 2025 – Feb. 2026
- Developed three CVAE-based surrogate models reconstructing velocity fields around a Voith propeller from high-fidelity CFD data - Integrated the trained surrogates into the in-house RANS solver FreSCo+, enabling drag prediction with propeller-hull interaction far faster than full CFD while retaining above 95% accuracy - Built fully automated Python pipelines covering simulation execution, data extraction, training, and validation	
M.Sc. Research Sharif University of Technology	Sep. 2019 – Mar. 2021
- Developed a coupled STAR-CCM+/ABAQUS framework for FSI problems, handling interface data transfer, mesh deformation, and coupled-solution convergence - Implemented inverse finite element solvers from first principles in MATLAB (stiffness assembly, boundary conditions) for real-time structural health monitoring	

Working Experience

Quality Control Engineer Shipyard — aluminium newbuild construction	Sep. 2023 – Sep. 2024
- Performed dimensional checks, plate thickness measurements, and weld inspections during hull assembly of new aluminium vessels, verifying compliance with class drawings and construction standards - Monitored structural assembly, coordinated with production teams on non-conformities, and maintained inspection records for project documentation	
Ship Structure Inspector	Sep. 2022 – Sep. 2023

- Conducted structural inspections of ship components including steel plate thickness measurements and assessment of structural condition
- Performed thickness measurements to identify corrosion, wastage, and areas requiring repair or replacement
- Carried out visual and surface inspections of welded and structural components, comparing results against design and allowable thickness requirements
- Documented inspection findings and supported preparation of technical inspection reports

Teaching Experience

Teaching Assistant Sharif University of Technology

2019 – 2020

↪ Course: Ship Structures

- Assisted in delivering course content and supporting student learning

Teaching Assistant Persian Gulf University

2016 – 2017

↪ Courses: Fluid Dynamics, Ship Design, Mechanical Vibrations

- Supported course instruction and mentored undergraduate students

Awards & Honors

Erasmus Mundus Scholarship Award Fully funded scholarship awarded on the basis of academic excellence and research potential, covering 18 months of the SEAS 4.0 double Master's programme **Sep. 2024**

1st Rank in B.Sc. Naval Architecture Top-ranked student throughout the Bachelor's program

2014 – 2018

Publications

Hybrid Negative-to-Positive Deadrise Hulls: A Performance Assessment for High-Speed Planing Vessels **2026**

E. Tahmasebi Boldaji, et al., Ocean Engineering ([access](#))

The Effects of Hydrodynamic Load on the Heave and Pitch Coupling Oscillations of a Plate **2024**

E. Tahmasebi Boldaji, et al., Journal of Marine Science and Technology ([access](#))

Engineering Related Projects

Explorer Yacht – Full Concept Design

↪ Tools: Maxsurf, Rhino, SolidWorks | Domain: Ship Design, Stability, Resistance, Seakeeping

- Executed a complete concept design cycle from market research and database analysis through to mission/route definition, general arrangement, and weight optimisation
- Performed hydrostatics, first-generation intact stability (IS-code compliance), and floodable length calculations
- Conducted resistance prediction, seakeeping natural period estimation, electrical load analysis, and machinery/propulsion selection
- Produced structural element sizing and a final 3D rendered concept model

CFD Resistance Analysis – Container Ship Hull with Roughness

↪ Tools: OpenFOAM, Star-CCM+ | Domain: CFD, Hydrodynamics

- Set up and executed RANS CFD simulations to predict ship hull drag at the design speed, considering the effect of surface roughness on drag increment.
- Performed mesh generation using snappyHexMesh and CFMesh, turbulence model selection ($k-\omega$ SST), and free-surface modelling (VOF method)
- Post-processed pressure and viscous drag components; validated results against ITTC resistance estimation methods



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